

EKRAMUL ALAM

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SUMMARY

CSE graduate with hands-on Android development experience building native Java applications with Firebase, MVVM architecture, and applied design patterns (Singleton, Factory, Observer, Adapter), backed by JUnit/Espresso testing and GitHub Actions CI. Strong foundations in Data Structures & Algorithms (Max Codeforces Rating: 1257), OOP, and RESTful API integration connecting mobile clients to back-end services. Also experienced in cross-platform Flutter and full-stack web development (MERN, Laravel). Known for clean coding practices, quick adaptability, and the ability to communicate technical concepts clearly and effectively.

EDUCATION

Khulna University of Engineering & Technology
Bachelor of Science in Computer Science and Engineering

Khulna, Bangladesh
Mar 2022 – Jul 2026

TECHNICAL SKILLS

Languages: Java, Kotlin, C++, Python, JavaScript, PHP

Mobile: Android SDK (Java), Firebase (Auth, Firestore, Cloud Storage), Volley, Gradle (Kotlin DSL), MVVM, ViewBinding, Navigation Component, Flutter

Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS

Web & Backend: React.js, Next.js, Node.js, Express.js, FastAPI, Laravel 10, REST APIs

Databases: Firebase Firestore, MongoDB, MySQL, PostgreSQL

Testing & DevOps: JUnit, Mockito, Robolectric, Espresso, Git, GitHub Actions CI/CD, Linux/Bash

Communication Skills: English (Full professional and Academic proficiency), Bengali (Native proficiency)

EXPERIENCE

Full-Stack Web Developer (Part-time)

Aug 2025 – Present

TechnoPLUS IT – Remote (Aus)

- Built and maintained a Laravel 9 + Vue.js LMS using Porto/Apiato modular architecture, applying OOP and design patterns (Repository, Service, Observer) across 15+ domain containers
- Independently researched and implemented a custom H5P interactive content integration from scratch, and designed RESTful APIs with role-based access control
- Implemented relational database tables with ordered pivot tables, SoftDeletes, and JSON metadata columns to support a flexible multi-content-type lesson structure

PROJECT EXPERIENCE

PharmaPlus – E-Commerce Pharmacy App *(Android, Java, Firebase)*

2024

- Built a dual-role (customer + admin) native Android e-commerce app in Java with Firebase Authentication, Cloud Firestore, and Cloud Storage, delivering product catalog, cart, multi-address checkout, order history, and an admin catalog-management panel
- Architected the app with MVVM (LiveData/ViewModel, ViewBinding, Navigation Component) and applied Singleton, Factory, Observer, and Adapter design patterns; consumed REST endpoints via Volley to connect the app to back-end services
- Wrote unit and instrumentation tests (JUnit, Mockito, Robolectric, Espresso) and automated builds through a GitHub Actions CI pipeline (Gradle Kotlin DSL, minSdk 28 / targetSdk 34)

Edge Intelligent SmartCane – Assistive App for the Visually Impaired *(Flutter, Kotlin, YOLOv8)*

2025 – Present

- Developed a cross-platform (Android/iOS) Flutter app that runs real-time on-device object detection through the device camera to assist blind and visually impaired users during navigation
- Integrated a fine-tuned YOLOv8 detection model with native Android (Kotlin) and C++ components, converting detections into audio alerts and haptic feedback for accessible, hands-free guidance

Automated Healthcare System *(REST API Backend, MERN)*

Nov 2024

- Designed and built RESTful APIs with a modular MVC backend (Node.js, Express, MongoDB) powering doctor appointment booking, pharmacy cart, blood donor search, and emergency ambulance services – the class of back-end services a mobile client consumes
- Implemented JWT-based authentication with bcrypt password hashing and role-based access control (Admin, Moderator, User), plus Cloudinary image storage and Nodemailer transactional emails

AI-Powered Learning Platform *(Next.js, FastAPI, LangChain)*

Jan 2026

- Developed a university course assistant with AI-powered tutoring and intelligent content search using LangChain and vector embeddings
- Engineered a FastAPI backend with Supabase (PostgreSQL) and row-level security, supporting real-time updates and separate role-based portals for students and instructors

ACTIVITIES

BUET CSE Fest 2026 Hackathon – Finalist | Dhaka, Bangladesh | Jan 2026

BITFEST 2025 – Executive Member | Khulna, Bangladesh | Dec 2024 – Jan 2025

SGIPC (Special Group of Interest in Programming Contest) – Campaign Editor | Khulna, Bangladesh | Aug 2023 – Present

Intra KUET Programming Contest 2023 – Best Team 2k20 Award Winner | Khulna, Bangladesh | 2024

Inter-University Programming Contests (IUPC) – Participant | KUET | 2025